Digital Table: Saving Data Forever

Python & CSV | Session 1

Welcome, Data Detective! Have you ever noticed that when you restart your Python program, your variables "forget" everything? Variables are like chalk on a blackboard—they are wiped clean easily!

The Memory Problem vs. Permanent Storage

Variables live in "Temporary Memory." When the computer stops, your data disappears like magic! Writing data to a file is like writing in a notebook. Even after the program ends, the "notebook" stays safe. Today, we use CSV files to remember everything!

What is a CSV File?

CSV stands for **"Comma Separated Values"**. It is the most common way for computers to share table-like data. Think of a CSV file as a **digital tiffin box**! Just like your tiffin has compartments separated by walls, a CSV file has data separated by commas.

```
Rahul, 10, Mumbai
```

Each comma creates a new compartment for your information! CSV files look just like your school timetable:

- **Rows:** Horizontal lines going across (like each day).
- **Columns:** Vertical sections going down (like each school period).

Reading CSV Files in Python

Python has a built-in helper called the `csv` module. We use the `with open` command to open our files safely. It makes sure the box is closed properly when we are done!

```
import csv
with open('marks.csv') as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

This code will print every row of your table as a Python List! When Python reads a row, each column has an **Index Number** starting from **0**.

The Data Detective Challenge!

Practice Worksheet | Grade 4 Coders

Test your detective powers on Python and CSV files! Read each question carefully and circle the best answer or fill in the blank lines. No cheating!

Question 1

What does the abbreviation **CSV** stand for?

A) Computer Smart Version

B) Comma Separated Values

C) Cookies, Soda, Vegetables

D) Correct System Variable

Question 2

Variables are like chalk on a blackboard because they live in what type of memory?

A) Permanent Memory

B) Temporary Memory

C) Forever Storage

D) Hard Drive Memory

Question 3

In our lesson analogy, a CSV file is like a digital _______ box where commas act as the compartment walls!

Question 4

When you look at a table, what do we call the **horizontal** lines going straight across?

A) Columns

B) Blocks

C) Rows

D) Commas

Question 5

When counting column index numbers in a Python list row, what number do we **always start counting from**?

A) 1

B) 0

C) -1

D) 10

Question 6

If Python reads the row: `['Rahul', '10', 'Mumbai']`, which code will correctly pick out the city name **'Mumbai'**?

A) row[0]

B) row[1]

C) row[2]

D) row[3]

Question 7

Which built-in Python block ensures that our file is safely and properly closed when we are finished with it?

A) with open()

B) close_now()

C) exit()

D) save_file()

Question 8

What is the name of the built-in module we must **import** at the very top of our code to work with spreadsheet data tables?

Question 9

True or False: If you open a CSV file in Notepad, you will see actual graphic borders and lines separating the words.

A) True

B) False

Question 10

Why do programmers love CSV files?

Question 11

In Python, why do we need to use `int(row[1])` before comparing numbers like marks?

Question 12

When you view a CSV file inside Microsoft Excel or Google Sheets, what happens to the commas?

A) They get deleted completely.

B) They turn into neat grid lines and cells.

C) They turn into question marks.

D) Nothing happens.